

NetraJyoti AI

A Bengali-first, safety-guidance and eye-care navigation MVP for rural West Bengal.

Submission information	Value
Team name	NETRAJYOTI
Team members and student IDs	Rahul Kar (bitsom_pm_2601248) Subhra Kundu (bitsom_pm_2601245)
Course	Product Management with Generative and Agentic AI
Phase	Phase 2
Product	NetraJyoti AI MVP

1. Product strategy and solution design

1.1 Problem statement

In rural West Bengal, Bengali-speaking users may not know whether an eye problem is serious, where to seek help, or which advice to trust. Informal advice, forwarded messages, cost, travel, waiting time, low digital confidence, and communication barriers can delay access to appropriate eye care. NetraJyoti provides Bengali-first, non-diagnostic safe next-step guidance and care navigation.

1.2 Design-thinking process

Stage	Application to NetraJyoti	Output
Empathise	Interview rural users, caregivers, ASHA workers, and outreach coordinators about eye-care decisions, language needs, and service barriers.	Evidence-backed needs and quotes.
Define	Frame the problem as delay, confusion, and lack of trusted Bengali guidance before care.	Problem statement and jobs-to-be-done.
Ideate	Explore voice, text, human-support, service-navigation, family-sharing, and safety-routing concepts.	Prioritised solution ideas.
Prototype	Create Bengali-first wireframes with urgent, routine, and human-support paths.	Interactive wireframe and user flow.
Test	Run task-based usability and content-comprehension sessions; test safety rules and fallbacks.	Insights, refinements, and release gates.

1.3 Personas and jobs-to-be-done

Persona	Need / job	Design implication
Maya - rural Bengali-speaking user	Understand what to do next when an eye concern creates fear or uncertainty.	Bengali-first, simple choices, large actions, no diagnosis claim.
Rahim - caregiver	Help a family member reach the right eye-care service and communicate urgency.	Selected symptoms, family message, share/copy, service direction.
Soma -	Support a resident who needs trusted local	Human-support route, PHC/ASHA options, clear

Persona	Need / job	Design implication
ASHA/community-health worker	guidance.	approved language.
Outreach coordinator	Keep camps, vision centres, clinics, and referral details current.	Governed service-directory update process and verified-at information.

1.4 User stories, MVP, and prioritization

User story / safety requirement	Related delivery scope
As a Bengali-speaking user, I want to choose my eye symptoms in Bengali so that I can understand a safe next step.	Consent, fixed symptom choices, urgent warning-sign check, deterministic routing.
As a caregiver, I want to share selected symptoms and approved guidance so that my family can act together.	Caregiver summary, share/copy fallback, service-direction support.
As an ASHA or community-health worker, I want a reviewed human-support route so that I can assist a resident safely.	Human-support path, PHC/ASHA guidance, clear escalation actions.
As an outreach coordinator, I want to update verified service details so that users receive current eye-care information.	Service-directory governance, verification ownership, data freshness.
Safety requirement: Deterministic reviewed rules must set the route before any AI summary is generated, so AI cannot change safety decisions.	Rules-first routing, controlled-AI feature flag, validation, approved fallback.

Priority	MVP capability	Why
Must	Bengali consent, fixed symptom selection, urgent warning check, deterministic reviewed route.	Core safety and comprehension.
Must	Urgent care advice, routine guidance, human-support path, return/restart actions.	No-dead-end safe navigation.
Must	District/town or block service direction, PHC/ASHA options, family sharing.	Transforms guidance into practical action.
Should	Optional browser voice input, listening support, patient-detail message consent.	Improves accessibility without blocking the flow.
Should	Controlled AI caregiver summary behind a feature flag and fallback.	Adds clarity only after deterministic route selection.
Could	Live service availability, richer partner integrations, appointment booking.	Post-MVP; needs operational data governance.

1.5 Build approach, testing, and AI

- Design and core development are handled internally to retain product knowledge, Bengali UX control, safety-rule ownership, and fast iteration. Specialist clinical/content review and local service-data verification are partner-supported.
- Technology capability: React + TypeScript + Vite frontend/PWA; Java 21 + Spring Boot REST API; PostgreSQL; Spring Security; JUnit/Mockito; React Testing Library and Playwright.
- Test before deployment through safety-rule unit tests, API integration tests, bilingual usability sessions, mobile/slow-network checks, voice/location denial tests, full urgent/routine/human journey tests, and content/clinical review.
- AI/GenAI is controlled and backend-only: deterministic rules decide urgency; AI is optional routine caregiver summarisation; an Enable AI Summary flag, policy/schema validation, timeouts, and approved Bengali fallback protect the safety boundary.

- Agentic AI is not allowed to autonomously route, diagnose, prescribe, or contact services. Future agent assistance may be used only for governed coordinator workflows such as service-data update validation and draft content preparation.

1.6 Product Strategy, Delivery and Technology Approach

Delivery Model: Internal Ownership and Selective Outsourcing NetraJyoti will retain product strategy, Bengali conversation design, safety rules, referral logic, privacy, quality assurance, and core application development internally. These are the elements that determine user trust and safety. We will selectively outsource commodity or specialist support such as cloud penetration testing, speech infrastructure, translation validation, and pilot recruitment. External vendors must not control clinical guidance or routing decisions.
Organisational Capabilities and Technology Choices NetraJyoti has full-stack web capability, Bengali-first UX/content capability, API and data modelling capability, automated-test capability, and product-safety governance. These support a React + TypeScript PWA for Android-friendly access; Java 21 + Spring Boot REST APIs for controlled backend logic; PostgreSQL for governed service data; Spring Security for API protection; and JUnit, Mockito, React Testing Library, and Playwright for quality assurance.
MVP Feature Scope The MVP includes Bengali consent, fixed symptom choices, optional voice with text fallback, urgent warning-sign check, deterministic URGENT/ROUTINE/HUMAN SUPPORT routing, approved Bengali guidance, district/town or block service direction, PHC/ASHA support, caregiver share/copy, privacy-safe feedback, and demo-labelled directory content. Controlled AI remains pilot-only and off by default.
Feature Prioritisation Used MoSCoW. Must: consent, fixed safety routing, urgent/routine/human guidance, accessible Bengali fallback, human support, service direction, safe family sharing, privacy, and feedback. Should: voice support and controlled-AI caregiver summary, which remains feature-flagged and off by default. Could: richer partner integrations, offline cache, reminders, and appointment support. Will not have in MVP: diagnosis, prescriptions, payment, medical records, or autonomous booking.
Pre-Deployment Testing and Release Readiness Tested every reviewed safety rule, API response, frontend journey, denied permission, poor-connectivity, browser/voice fallback, share/copy fallback, data-flow boundary, and AI failure mode. Conducted clinician/content review, Bengali usability sessions, teach-back comprehension checks, accessibility checks, service-data verification, privacy tests, and an incident/release-gate review before any pilot.
AI, GenAI, and Agentic AI Integration NetraJyoti uses AI-enabled Bengali speech input and output to improve accessibility. Safety routing remains fully deterministic: reviewed rules determine the URGENT, ROUTINE, or HUMAN-SUPPORT path before any GenAI capability is used. GenAI is limited to an optional plain-Bengali explanation or caregiver summary based only on approved, route-specific content. It cannot diagnose, prescribe, change urgency, or select a service. The backend feature flag is off by default; policy and schema validation reject prohibited output; timeouts or failures immediately show approved Bengali fallback content. No patient-identifiable information is sent to an AI provider. Agentic AI is future scope only. It may support governed coordinator tasks, such as validating service-directory updates or preparing content drafts for human review. It will never autonomously route users, diagnose, prescribe, contact services, or publish unverified service information.

2. Experimental task A: five JTBD interviews

Required evidence structure

Conduct five real interviews based on the main problem statement. Use a JTBD-style script that explores a real recent eye-care decision, current workaround, anxiety/friction, desired outcome, trusted information source, and reasons for delay. Record the interviews only with participant consent.

Participant	Exact quote	Insight	Feature mapping	Recording link
P1	"I don't know if this is serious"	Users need safe urgency clarity	Deterministic urgent/routine/human-support routing	https://drive.google.com/file/d/1b7s9hYg9FeA5tmZaSeYt5VlzAJmELrK/view?usp=drive_link
P2	"I can't type/read English"	Language and literacy block access	Bengali voice input, recognised-text review, text/fixed-choice fallback	https://drive.google.com/file/d/1Xcj8n3xReNfcPQ4wZW-uPEcSlmUlfUUQ/view?usp=drive_link
P3	"I don't know where to go"	Guidance must become practical navigation	District/block service direction, PHC/ASHA options	https://drive.google.com/file/d/1pg8SybVlhFRRh3HVhUqw6e96TDsaFGxd/view?usp=drive_link
P4	"I need to ask other family members"	Care decisions are shared	Caregiver message, browser share/copy fallback	https://drive.google.com/file/d/12f8oSr1wpeM1OH-5R4F1o05B9FfuP7t0/view?usp=drive_link
P5	"I'm not comfortable using this alone"	Human reassurance is essential	ASHA/PHC/eye-clinic/trusted-family human-support path	https://drive.google.com/file/d/1_IKpLxz2mwhgLP_D_2WxUBTs-51eJTdU/view?usp=drive_link

3. Delivery, governance, Jira, and Lean Canvas

3.1 Methodology and project plan

NetraJyoti uses Lean UX plus Agile Kanban. Lean UX reduces product uncertainty through real user evidence and rapid Bengali prototype tests. The Kanban board visualises the flow of reviewed requirements into testable increments, while safety and content review act as defined release gates.

Phase	Weeks	Deliverable / gate
Discovery and evidence	1-2	JTBD interviews, personas, Lean Canvas, hypotheses.
Journey and prototype	3-4	User flow, bilingual wireframes, usability findings.
Technical design and backlog	5	PRD, HLD, Jira stories, estimates, safety rules.
MVP build	6-9	Core routing; urgent/routine/human routes; service direction; sharing; controlled AI/fallback.

Phase	Weeks	Deliverable / gate
Pilot readiness	10	Privacy, content, service-data, and release checks.
Validation and final pack	11-12	Usability/pilot insight, refinements, final artefacts and demo.

3.2 Jira delivery evidence

Epic: [NET-1 - NetraJyoti Epic MVP](#)

Spaces / NetraJyoti / NET-1

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance MVP

Add a child work item

Link work item

Create

...

Key details

Rahul Kar raised this request via Jira

Description

Problem

Rural Bengali-speaking users may delay eye-care decisions because they are unsure how serious a concern is, where to go, and which advice to trust. Text-heavy or English-first services create additional barriers.

Feature Goal

Provide a safe Bengali voice-first experience that helps users understand the appropriate next step and receive reliable service direction. NetraJyoti is guidance and referral support—not a diagnostic tool—and users must verify service availability locally before travel.

Target Users

- Rural Bengali-speaking users
- Caregivers
- ASHA/community-health workers
- Eye-care outreach coordinators

Kanban board: [Board 68](#)

NetraJyoti AI MVP Delivery Board

Kanban board

Reports

+

Q Search board

Version Epic Label

Quick filters

Group: Query

TO DO

IN PROGRESS

TESTING / CONTENT REVIEW

DONE 11

Story: AI-assisted caregiver summary from approved result content

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-12

5

Story: AI-assisted plain-Bengali explanation of approved guidance

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-H

5

Story: Feedback and privacy-safe pilot analytics

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-10

8

Story: Safety, accessibility and fallback quality assurance

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidanc...

NET-9

8

Story: Consent, privacy and minimum-data handling

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidanc...

NET-8

8

Story: As an outreach coordinator, I want to update

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NetraJyoti AI MVP Delivery Board

Kanban board

Reports

Search board

NET-7

NET-6

NET-5

NET-4

Version

Epic

Label

Quick filters

Group: Query

TO DO

IN PROGRESS

TESTING / CONTENT REVIEW

DONE 11

Story: As an outreach coordinator, I want to update eye-camp dates and locations so users receive current service information

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-7

Story: As an ASHA worker, I want an assisted flow so I can guide a villager safely without acting as a clinician

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-6

Story : As a caregiver, I want a short audio referral summary so I can help my family member travel and prepare

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-6

Story: As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-4

NetraJyoti AI MVP Delivery Board

Kanban board

Reports

Search board

NET-6

NET-5

NET-4

NET-3

Version

Epic

Label

Quick filters

Group: Query

TO DO

IN PROGRESS

TESTING / CONTENT REVIEW

DONE 11

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-6

Story : As a caregiver, I want a short audio referral summary so I can help my family member travel and prepare

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-5

Story: As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

NET-4

Story: As a user with sudden vision loss, severe pain, eye injury or chemical exposure, I want immediate advice to seek emergency care

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance...

NET-3

Story: As a rural Bengali-speaking user, I want to describe my eye problem by voice so I do not need to type or read English

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidanc...

Dashboard: [Dashboard 10035](#)

NetraJyoti AI MVP Dashboard V1

Created vs Resolved Chart: NetraJyoti Epic MVP

10

8

6

4

2

0

05 Jul 2026

09 Jul 2026

14 Jul 2026

19 Jul 2026

23 Jul 2026

28 Jul 2026

01 Aug 2026

Issues in the last 30 days (grouped daily)

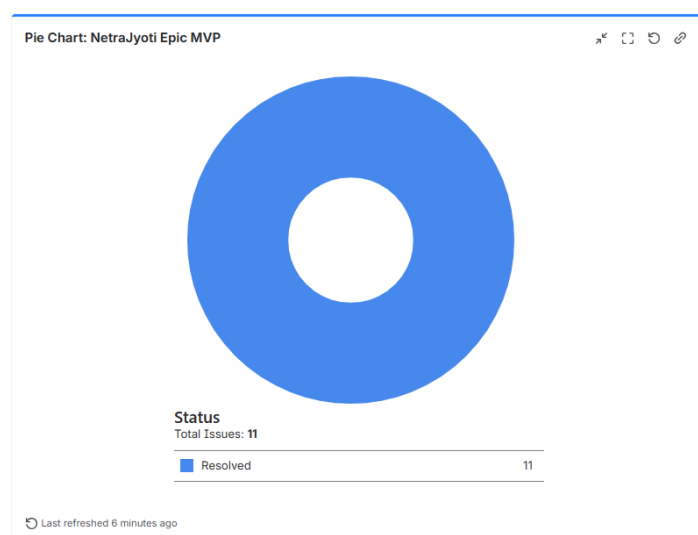
View in Issue navigator

Created issues (11)

Resolved issues (11)

Last refreshed 5 minutes ago

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Filter Results: NetraJyoti Epic MVP

T	Key	Summary	P
☒	NET-12	Story: AI-assisted caregiver summary from approved result content	=
☒	NET-11	Story: AI-assisted plain-Bengali explanation of approved guidance	=
☒	NET-10	Story: Feedback and privacy-safe pilot analytics	=
☒	NET-9	Story: Safety, accessibility and fallback quality assurance	=
☒	NET-8	Story: Consent, privacy and minimum-data handling	=
☒	NET-7	Story: As an outreach coordinator, I want to update eye-camp dates and locations so users receive current service information	=
☒	NET-6	Story: As an ASHA worker, I want an assisted flow so I can guide a villager safely without acting as a clinician	=
☒	NET-5	Story : As a caregiver, I want a short audio referral summary so I can help my family member travel and prepare	^
☒	NET-4	Story: As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease	^
☒	NET-3	Story: As a user with sudden vision loss, severe pain, eye injury or chemical exposure, I want immediate advice to seek emergency care	^

1-10 of 11

Last refreshed 7 minutes ago

3.3 Ethical considerations

Area	Risk	Safeguard
Non-diagnosis	False clinical confidence.	Clear purpose; no disease label, prescription, or clinician-replacement claim.
Urgency	Delay or inappropriate escalation.	Versioned clinician-reviewed deterministic rules; rule tests and approved urgent content.
AI	Invented advice or changed urgency.	Backend-only feature flag; routine-only; schema/policy validation; immediate approved fallback.
Privacy	Exposure of name, phone, address, voice, or location.	Data minimisation, explicit consent, client-side temporary details, no PII logs by default.
Inclusion	Language, literacy, device, age, disability, or connectivity exclusion.	Bengali-first UX, large targets, optional voice, typed/fixed-choice fallback, human support.
Service data	Incorrect camp/clinic information.	Demo/verified-at labels, outreach ownership, update cadence, no unsupported availability claims.

3.4 Lean Canvas

Block	NetraJyoti response
Problem	Confusion, delay, mistrust, and service uncertainty before eye care.
Customer segments	Rural Bengali-speaking users; caregivers; ASHA/community workers; outreach coordinators.
Unique value proposition	Bengali-first safe next-step guidance for eye problems.
Solution	Reviewed safety routing, service direction, human support, and family sharing.
Channels	Eye-care partners, eye camps, ASHA workers, QR/posters, caregiver sharing, and local outreach.
Costs	Product build, cloud/data, clinical/content review, service verification, usability research, and optional AI.
Key metrics	Journey completion, route distribution, service-direction use, comprehension, share rate, data freshness, AI fallback rate.
Unfair advantage	Local partner trust, reviewed Bengali content, safety workflow, and curated service information.

4. STP, brand, 7Ms, and launch plan

4.1 STP and brand

Element	NetraJyoti plan
Segmentation	Rural/near-rural Bengali speakers; eye-care uncertainty level; caregiver role; digital confidence; proximity to vision/eye-care services.
Targeting	Primary: Bengali-speaking adults and caregivers in rural West Bengal with a current eye concern and uncertainty about the appropriate next step.
Positioning	NetraJyoti helps people understand a safe next step and the suitable path to eye care in Bengali. It does not diagnose disease.
Brand	Name: NetraJyoti AI. Meaning: light/clarity for the eyes. Personality: calm, respectful, practical, trustworthy, Bengali-first.
Brand promise	Clear, safe next steps for eye-care decisions, in the user's language.

4.2 7Ms for the new product

M	Decision
Mission	Reduce delay, fear, and confusion before appropriate eye care.
Market	Rural West Bengal communities reached through care partners and community-health networks.
Message	Bengali-first safe next-step guidance; not a diagnosis.
Media	ASHA/partner referrals, eye-camp posters, QR cards, WhatsApp family sharing, local outreach, and partner channels.
Money	Pilot funded through CSR, NGO, eye-care partners, public-health grants, or implementation agreements rather than user payment.
Manpower	Product, engineering, QA, clinical/content reviewer, outreach coordinator, partner health workers.
Measurement	Completion, route/service use, comprehension, verified data freshness, pilot feedback, and safety/fallback signals.

4.3 End-to-end launch plan

Launch phase	Actions	Exit gate
Prepare	Approve safety/content, verify service data, train partner/ASHA referrals, create QR/outreach materials, test live MVP.	Clinical, privacy, usability, and operational approval.
Pilot launch	Launch with one or two partner areas; use assisted onboarding at camps/vision centres; capture non-identifying journey evidence.	No critical safety or comprehension failure.
Learn	Review friction, route distributions, data freshness, share use, and interviews weekly; prioritise fixes.	Evidence supports a wider pilot.
Scale carefully	Expand partner locations after service-data ownership and support model are proven.	Operational readiness and documented impact hypothesis.
60-second launch pitch. NetraJyoti AI is a Bengali-first eye-care guidance service for people who are unsure what to do when an eye problem occurs. It does not diagnose disease. Instead, it helps users identify warning signs, understand a safe next step, and find the right eye-care route such as a vision centre, clinic, hospital, PHC, or trusted health worker. We will launch through eye-care partners, ASHA/community workers, eye camps, and simple mobile access so guidance reaches people before uncertainty becomes delay.		

5. Channel strategy, user flow, and channel experiment

5.1 Omnichannel strategy

NetraJyoti should be omnichannel, not purely online. A mobile web/PWA experience provides scalable, low-friction access, while eye camps, vision centres, ASHA workers, PHCs, and trusted caregivers provide human reassurance and support for people with low digital confidence or limited connectivity.

Channel	Role	Owned / partner	Parity control
Mobile PWA	Self-service symptom, route, service direction, family message.	Owned digital experience.	Same approved rules/content version and service-directory data.
ASHA / PHC / eye-care partner	Assisted onboarding, human escalation, referral, service confirmation.	Partner/franchised-like local delivery.	Training, approved scripts, referral standards, data-update process.
Eye camps / vision centres	Physical awareness, QR entry, service follow-through, feedback.	Partner channel.	Consistent brand, printed scripts, verified service details.
Family sharing / WhatsApp	Caregiver involvement and urgency communication.	User-owned social channel.	Approved share template retains safe next step and selected symptoms.

NetraJyoti AI - Two-Channel Experiment

1. Objective

Compare the same non-diagnostic eye-care enquiry in two channels to identify completion time, friction, drop-offs, trust leakage, and scaling constraints.

2. Test scenario

A Bengali-speaking caregiver wants guidance for an older family member with red, itchy, watering eyes and is unsure whether to go to a vision centre, PHC, or eye clinic.

Success criterion: the caregiver reaches a safe routine-care recommendation and receives a shareable care summary or assisted referral direction without being shown a disease diagnosis.

3. Channels tested

Channel	Real action attempted	Test setup
PWA self-service	Open NetraJyoti, give consent, select symptoms, complete the warning-sign check, select district/town, view routine guidance, and create a caregiver summary.	Android phone, mobile data, Bengali-first UI.
Assisted partner enquiry	Ask an ASHA worker or eye-camp outreach coordinator for the same next-step guidance and nearby service direction.	In-person or phone-assisted enquiry using the same scenario.

4. Observation log

Measure	PWA self-service	Assisted partner enquiry
Participant label	PWA-01 (caregiver, 38)	Assisted-01 (caregiver, 41)
Time to begin	25 seconds	2 minutes 10 seconds

Measure	PWA self-service	Assisted partner enquiry
Time to safe next step	3 minutes 42 seconds	8 minutes 35 seconds
Time to share / service direction	4 minutes 15 seconds	10 minutes 05 seconds
Outcome	Routine eye-check recommendation and shareable summary created.	PHC / vision-centre direction explained verbally; no written summary initially.
Completion	Completed independently after one prompt.	Completed with staff availability.

5. Friction points and drop-off moments

Channel	Observed friction	Potential drop-off	Recommended improvement
PWA	Consent instruction was read but the checkbox was initially missed.	User may leave before starting.	Place a short Bengali instruction directly above a large consent control.
PWA	District/town selection depends on familiarity with administrative names.	User may exit when locality is not easily recognised.	Add searchable Bengali locality names, recent choices, and support fallback.
PWA	User hesitated before allowing location access.	User may reject permission and abandon service search.	Explain location is optional; offer district/town selection first.
Assisted	Worker was not immediately available.	User may delay until next visit or give up.	Provide QR/PWA self-service and escalation contact path.
Assisted	Guidance was verbal and details were not retained.	Family may forget suggested service or urgency message.	Generate a Bengali shareable summary the worker can send or read aloud.

6. Trust leakage and scaling constraints

Area	Finding	Why it matters	Mitigation
PWA trust	User questioned whether an app can understand an eye problem.	Fear of an incorrect diagnosis can prevent use.	Explain value positively: safe next steps and a suitable eye-care route; keep non-diagnostic boundary concise.
Directory trust	Service listings may look unverified or outdated.	A wrong location damages trust quickly.	Show verification date and service type; label demo data; use coordinator update workflow.
Assisted trust	User depends on a worker's availability or memory.	Experience varies by person and day.	Use standard guidance cards and shareable summaries linked to reviewed rules.
PWA scale	Access, connectivity, voice/browser support, and digital confidence vary.	Low marginal cost but needs assisted onboarding and simple design.	Use an offline-friendly PWA and local-language support.
Assisted scale	ASHA/outreach availability and training are limited.	High trust but capacity grows slowly.	Use scripts, referral workflow, and service-directory governance.

7. Conclusion

The PWA route is faster when the user can operate a phone comfortably and identify their area. The assisted route creates more human trust and supports low-confidence users, but it is slower and depends on worker availability.

5.2 User flow and journey

NetraJyoti AI — simple user journey flow

Reviewed safety rules determine the route. Optional AI never determines urgency.

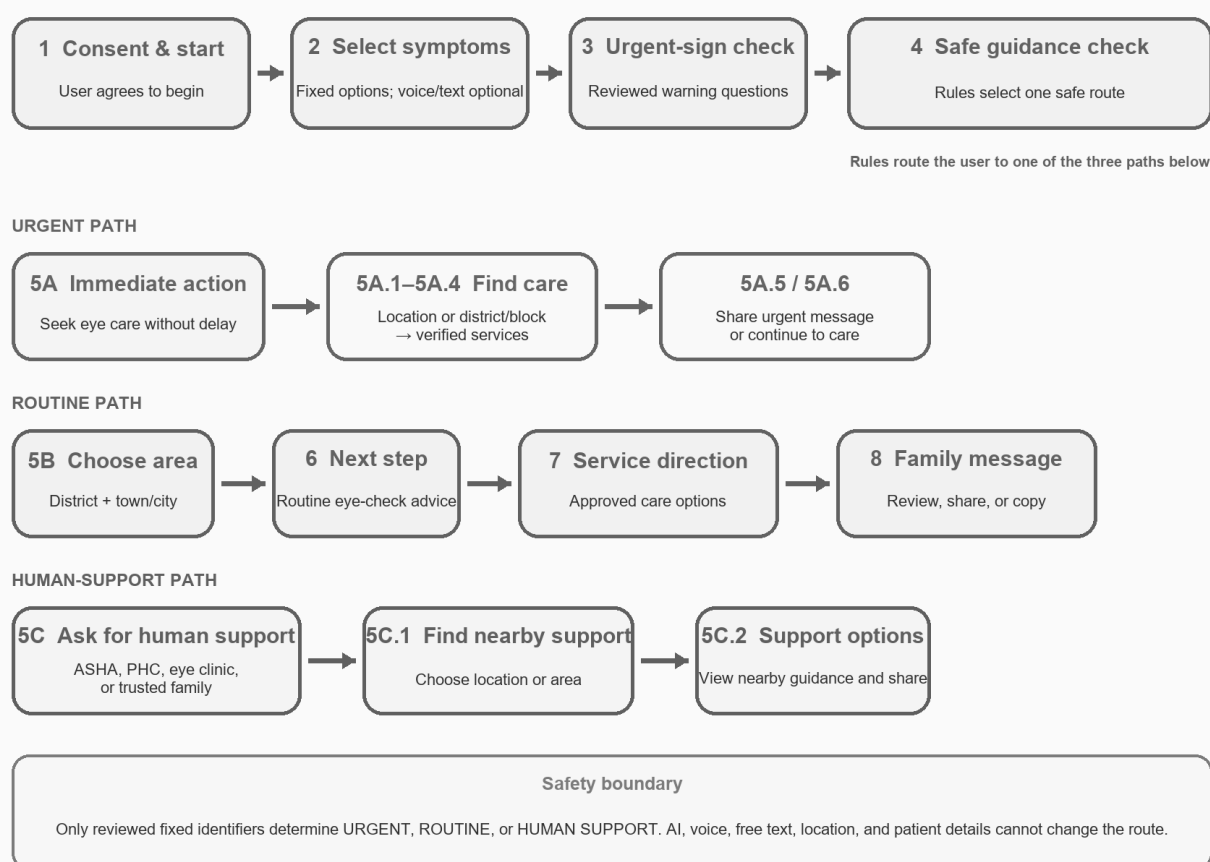


Figure 1. NetraJyoti AI end-to-end journey: core intake followed by urgent, routine, or human-support routes.

6. Retention, CLV, customer service, and growth

6.1 Retention and renewal research

Participant	Verbatim renewal objection	Emotional driver	Retention lever	Video link
R1	Concern that the product may not be trusted for an eye-care decision	Trust	Show clinician-reviewed safety boundary; add visible human-support option	https://drive.google.com/file/d/1iLTcu5Fd_LUWqA2mkzf6HhJC7-GhIAcF/view?usp=drive_link
R2	Concern that the process may be difficult for users with low digital confidence	Complexity	Bengali voice/text fallback, simple steps, and ASHA/PHC-assisted support	https://drive.google.com/file/d/1TuiJbZ_ZcpFi6-1wh647SC5LyVa30hs93/view?usp=drive_link
R3	Concern that the product may not be useful after the first guidance session	Low perceived value	Onboarding explaining repeat-use situations; short follow-up education/support	https://drive.google.com/file/d/1tJGtZTGYdTdY1-jYok5-AUs8xDKAH0wc/view?usp=drive_link
R4	Concern about accessibility of the app	Complexity, convenience	Simple steps, ASHA/PHC-assisted support	https://drive.google.com/file/d/1nqYhq4s_fz9oW9inAqAzLOEcVWDJLEVG/view?usp=drive_link
R5	Concern that the product might not be trusted and should incorporate notification and follow up messages	Trust	More information, human support option	https://drive.google.com/file/d/1IGCzljpkql7rkOwn8MGc25Drry_yAh1L/view?usp=drive_link

6.2 Customer-service and CLV strategy

Area	Approach
Customer support	In-app help, family-share guidance, ASHA/PHC/clinic escalation, clear service-data disclaimer, and feedback route. No automated clinical diagnosis.
Increase trust and repeat use	Keep content short and understandable; show safe next step; maintain verified service data; make caregiver support simple; offer human fallback.
CLV / renewal	For partner-funded model: demonstrate value through completed care-navigation journeys, service-direction use, community reach, and partner trust rather than consumer subscription revenue.
Growth loops	Caregiver share loop; partner/ASHA referral loop; eye-camp QR loop; service-data update loop; learning loop from anonymised friction and feedback.
Habit building	Use only ethically appropriate reminders/support after explicit consent. Avoid dark patterns, fear-based nudges, and health claims.

7. Experimental task C: business-owner support research

Business-owner support research: retain evaluator-accessible recordings and document five customer-support queries, best/worst resolution times, follow-ups, escalations, drop-offs, and specific frustration moments before final submission.				
Owner / industry	Frequent issue / friction	Current resolution	Improvement suggestion	Recording link
B1(Cloud Kitchen Owner)	Order delays during peak hours and incorrect orders affecting customer ratings.	Kitchen staff manually prioritize orders and resolve complaints through delivery platforms.	Implement order management software with real-time kitchen queue tracking, standardized preparation workflows, and packaging verification before dispatch.	https://drive.google.com/file/d/1HDBlyE2KTctfrqzGaYwRXxFR5i88DTJ6/view?usp=drive_link
B2 (Caregiver Center owner)	Difficulty managing schedules, medication reminders, and communication with multiple family members.	Uses paper notes, phone calls, and messaging apps to coordinate care.	Use a shared caregiving app for schedules, medication tracking, task updates, and family notifications to reduce missed tasks and improve transparency.	https://drive.google.com/file/d/1W5Of5-O0O5hWXY37TjplUDdTIAVvslqg/view?usp=drive_link
B3(Fruit seller)	Unsold fruits spoil quickly, leading to daily inventory loss and inconsistent income.	Seller reduces prices at the end of the day or absorbs the loss personally.	Introduce simple demand forecasting, digital inventory tracking, and partnerships with nearby shops or food vendors to sell excess stock before spoilage.	https://drive.google.com/file/d/144bs6sOx2cgXxvUj81dLSTPn2AU1iWqH/view?usp=drive_link

Application to Netrajyoti: expected support design

- Provide clear route explanations and visible next-step actions instead of generic advice.
- Offer human support and caregiver/family sharing alongside self-service guidance.
- Clearly state that service directions are guidance only; users should confirm availability before travelling.
- Track non-identifying friction signals and verify service information regularly.
- Use simple, respectful Bengali and provide a clear restart or return option at every major point in the journey.

8. Jira Epic backlog traceability

Current backlog status. NET-1 is In Progress. The Epic has 11 linked delivery stories (NET-2 to NET-12). All 11 stories are Resolved in the current Jira export, with 51 total story points. This is the source of truth for the delivery appendix.			
Jira story	Backlog scope	Priority / status / points	PRD and journey traceability
NET-2	Bengali voice capture with recognised-text review and text fallback.	High / Resolved / 5	Voice-first access; Step 2 symptom input; accessibility fallback.
NET-3	Immediate urgent-care advice for sudden vision loss, severe pain, injury, or chemical exposure.	High / Resolved / 5	Step 3 red-flag check; Step 4 urgent route; Step 5A branch.
NET-4	Plain Bengali, non-diagnostic next-step guidance for non-urgent concerns.	High / Resolved / 5	Routine outcome; Steps 5B, 6, 7, and 8.

NET-5	Caregiver audio/referral summary for travel and preparation.	High / Resolved / 5	Family-sharing flow; urgent 5A.5 and routine Step 8.
NET-6	ASHA-assisted journey for safely supporting a villager.	Medium / Resolved / 3	Human-support path; Steps 5C, 5C.1, and 5C.2.
NET-7	Outreach coordinator updates eye-camp dates and location information.	Medium / Resolved / 5	Verified service-directory governance and data freshness.
NET-8	Consent, privacy, and minimum-data handling.	Medium / Resolved / 5	Step 1 consent; optional details/share consent; data boundaries.
NET-9	Safety, accessibility, and fallback quality assurance.	Medium / Resolved / 5	Deterministic rules, tested fallbacks, usability/release gates.
NET-10	Feedback and privacy-safe pilot analytics.	Medium / Resolved / 5	Pilot metrics, non-content analytics, operational learning.
NET-11	Controlled AI plain-Bengali explanation of approved guidance.	Medium / Resolved / 5	Server feature flag; rules-first route; validation and fixed-copy fallback.
NET-12	Controlled AI caregiver summary from approved result content.	Medium / Resolved / 3	Optional Step 8 summary; no user-entered data sent to provider; fallback.

Evidence and Supporting Links

- High_Level_Design_Document: docs/project-documents/NetraJyoti_High_Level_Design.pdf
- Technical_PRD.Document: docs/project-documents/NetraJyoti_Technical_PRD.pdf
- PRD: docs/project-documents/NetraJyoti_MVP_PRD.pdf
- Project Plan: docs/project-documents/NetraJyoti_Project_Plan.pdf
- NetraJyoti wireframe: <https://rahulkarcj.github.io/netrajyoti-ai-mvp/>
- GitHub repository: <https://github.com/rahulkarcj/netrajyoti-ai-mvp>
- Jira Epic: <https://rkarpmdemo.atlassian.net/browse/NET-1>
- Jira dashboard: <https://rkarpmdemo.atlassian.net/jira/dashboards/10035>
- Jira Kanban board: <https://rkarpmdemo.atlassian.net/jira/people/557058:3da47173-80fe-4c5f-9db8-804c2541a6c8/boards/68>
- MVPDemoRecording: https://drive.google.com/drive/folders/1nOJ4W02Vaz4ZS5RFRv_ljrkiAqF0jnJA?usp=drive_link